ADVANCED PROGRAMMING WITH PROJECT-I/MINOR PROJECT REPORT

Entitled

A.L.E.R.T. (ADAPTIVE LANDMARK EYE-STATE & REACTION-TIME): A LANDMARK-BASED, FULLY OFFLINE DRIVER FATIGUE MONITOR

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CERTIFICATE

This is to certify that Kavya Kothari (2301030600045), Prem Deshani (2301030600043) and Mehulsinh Rathod (2301030600049) have submitted the Advanced Programming with Project-I/Minor Project report entitled A.L.E.R.T. (Adaptive Landmark Eye-state & Reaction-Time) in partial fulfillment of the requirement for the award of the degree in Bachelor of Technology in Computer Engineering (AI & ML) of Silver Oak University, Ahmedabad during the academic year 2026-2027.

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DECLARATION

We hereby declare that the Advanced Programming with Project-I/Minor Project report submitted along with the Project entitled A.L.E.R.T. (Adaptive Landmark Eye-state & Reaction-Time) in partial fulfillment for the Bachelor of Technology in Computer Engineering (AI & ML) to Silver Oak University, Ahmedabad, is a bonafide record of original work carried out by us at the College of Technology under the supervision of Ms. Vishwa Patel.

We further declare that the report does not contain any material previously submitted for the award of any degree or diploma in this or any other university. It is also certified that no part of this report has been copied from any source without proper acknowledgment, and all referenced materials, figures, and information from other sources have been appropriately cited in the report.

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ABSTRACT

Camera-based driver monitoring can warn of fatigue before a lapse becomes a crash, but cloud-hosted designs add latency, fail without coverage and send the driver's face off the vehicle. This project presents ALERT, a driver fatigue monitor that runs entirely on a consumer CPU. It reads facial landmarks from MediaPipe Face Landmarker and computes a nine-feature vector per frame: eye aspect ratio (EAR), its short-term deviation, a 60-second closure ratio, blink rate, blink duration, mouth aspect ratio (MAR), yawn count, head pitch and the ratio of MAR to EAR. A 10-second calibration sets driver-specific thresholds, a slow moving average tracks drift, and fatigue is graded on four levels from alert to microsleep.

The objectives are an offline pipeline from frames to a fatigue grade, per-session calibration, and a documented path to training on the UTA-RLDD dataset. A working proof of concept (features, calibration, Random Forest, live overlay) is complete, with its classifier trained only on synthetic data, so no real-world accuracy is claimed. A feature cache of 141 RLDD videos (84,942 windows) and a CPU training benchmark are complete. The planned two-layer bidirectional GRU with attention and a proposed LightGBM–GRU–CNN ensemble are yet to be trained on real footage.

Keywords: driver fatigue detection, eye aspect ratio, PERCLOS, MediaPipe, on-device inference, temporal models

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LIST OF ABBREVIATIONS

ALERT	Adaptive Landmark Eye-state & Reaction-Time
BiGRU	Bidirectional Gated Recurrent Unit
CLAHE	Contrast-Limited Adaptive Histogram Equalisation
CNN	Convolutional Neural Network
CPU	Central Processing Unit
DMS	Driver Monitoring System
EAR	Eye Aspect Ratio
FPS	Frames Per Second
MAR	Mouth Aspect Ratio
MOE	Mouth-Over-Eye Ratio (MAR/EAR)
NTHU-DDD	National Tsing Hua University Driver Drowsiness Detection Dataset
ONNX	Open Neural Network Exchange
PERCLOS	Percentage of Eye Closure
POC	Proof of Concept
PnP	Perspective-n-Point
RLDD	UTA Real-Life Drowsiness Dataset

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CHAPTER 1 INTRODUCTION

1.1 PROJECT SUMMARY

ALERT (Adaptive Landmark Eye-state & Reaction-Time) is a driver fatigue monitor that runs entirely on a consumer CPU. It reads facial landmarks from MediaPipe Face Landmarker, computes a nine-feature temporal vector per frame, calibrates driver-specific thresholds in 10 seconds, and grades fatigue on four levels from alert to microsleep. A proof of concept (POC) is complete; its classifier is trained only on synthetic data, so no real-world accuracy is claimed.

1.2 BACKGROUND AND MOTIVATION

NHTSA estimates that 91,000 police-reported crashes in 2017 involved a drowsy driver, with about 50,000 injuries and nearly 800 deaths, and cautions that the true figures are likely higher [1]. A 2024 AAA Foundation analysis estimated that 17.6% of U.S. fatal crashes from 2017 to 2021 involved a drowsy driver [2]. At 100 km/h a vehicle covers roughly 56 m in the two seconds a microsleep can last. Regulation (EU) 2019/2144 requires a driver drowsiness and attention warning in new vehicle types from July 2022 and in all newly registered vehicles from July 2024 [3].

1.3 PURPOSE

Cloud-hosted designs add network latency, fail without coverage and send the driver's face off the vehicle. ALERT aims to show that a low-cost, passive, single-camera, fully offline pipeline is feasible.

1.4 OBJECTIVES

- Build an offline pipeline from camera frames to a four-level fatigue grade.
- Calibrate EAR and MAR thresholds per session, with noise rejection and slow drift adaptation.
- Train a temporal model (BiGRU with attention) on the UTA-RLDD dataset using subject-independent folds.
- Achieve real-time operation on a CPU without a dedicated GPU.

1.5 SCOPE

In scope: landmark-based features, per-session calibration, the POC Random Forest, the production BiGRU model, a live overlay and tiered audio alerts. Out of scope: vehicle integration, haptic output, in-vehicle validation, night-time infrared operation and any safety-critical certification. ALERT is a research prototype and must not be relied on as a safety-critical system.

CHAPTER 2 LITERATURE REVIEW

2.1 TECHNOLOGY AND RELATED WORK

2.1.1 Eye Closure Measures

Soukupová and Čech defined the eye aspect ratio (EAR) from six landmarks per eye and detected blinks from its temporal pattern [5]. PERCLOS, the proportion of time in a minute that the eyes are at least 80% closed, was established in driving-simulator work [4] and validated against psychomotor vigilance performance [6], [7]. A 2023 review notes that definitions vary across studies and that PERCLOS is less reliable for moderate drowsiness, which argues for combining it with other signals [8].

2.1.2 Landmark Extraction

Kazemi and Sullivan's regression-tree method underlies the 68-point dlib predictor [9]. MediaPipe is a cross-platform framework for perception pipelines [10]; its face mesh infers 468 3D landmarks from one camera [11]. The current Face Landmarker task returns 478 landmarks, optional blendshapes and a facial transformation matrix [12].

2.1.3 Sequence Models

Gated recurrent units [17] and additive attention [18] are standard tools for temporal classification. LightGBM [19] is used for the tabular branch of the proposed ensemble.

2.2 DATASETS AND REPORTED RESULTS

Table 2.1 summarises the datasets and results relevant to this work.

Table 2.1 Datasets and Reported Results

Source	Data	Reported result
NTHU-DDD [13], [15]	Infrared, 36 subjects, acted drowsiness	73.06% (deep network)
UTA-RLDD [14]	About 30 h RGB, 60 subjects, natural labels	65.2% video-level (human observers 57.8%)
Xu et al. [16]	RLDD, hybrid features, soft-voting ensemble	86.52% video-level

RLDD was not recorded while driving and lacks night lighting and vehicle vibration [14]. The result in [16] supports per-driver thresholds and multi-region features, and shows that tree-based ensembles on hand-built features are competitive.

2.3 RESEARCH GAP

- Coarse output: many systems emit a binary label; ALERT grades four levels.
- Fixed thresholds: a population EAR threshold fits poorly when resting eye openness differs; ALERT calibrates per session.
- Single-channel fragility: landmark-only systems degrade under eyewear or low light; the proposed ensemble adds temporal and pixel channels (not yet measured).
- Connectivity and privacy: ALERT processes frames locally and persists only calibration thresholds.
- Data realism: neither NTHU-DDD nor RLDD represents in-cab conditions; RLDD is the primary source and the gap is an open problem.

CHAPTER 3 PROJECT PLANNING AND SCHEDULING

3.1 DEVELOPMENT APPROACH AND JUSTIFICATION

An incremental approach was used: first a threshold-based real-time baseline, then the data pipeline, then the deep-learning model. Landmark features were chosen over heavy 3D-CNNs because the target is CPU-only operation (guide-agreed focus, Month 1 review).

3.2 TEAM AND RESPONSIBILITIES

The group (Project ID CE_AIML_31) comprises Prem Deshani (2301030600043), Kavya Kothari (2301030600045) and Mehulsinh Rathod (2301030600049) under the guidance of Ms. Vishwa Patel. [Add each member's specific responsibilities.]

3.3 SCHEDULE AND PROGRESS

Table 3.1 Periodic Review Summary

Period	**Work completed**	**Progress**
Month 1 (08/08/2026 to 19/09/2026)	Landmark and feature extraction, calibrator, BiGRU model, live dashboard, audio alerts	80%
Week 1 (20/09/2026 to 26/09/2026)	Feature cache for 141 RLDD videos (84,942 windows), augmentation bug fix, CPU training benchmark, Review-2 material	90%

Guide meetings were held on 08/08/2026, 22/08/2026, 05/09/2026 and 19/09/2026. Planned next: 25-epoch BiGRU training on Fold 4 validation, per-class F1 scores and ONNX export.

CHAPTER 4 SYSTEM ANALYSIS

4.1 STUDY OF CURRENT SYSTEMS

Existing approaches use vehicle-dynamics signals (steering, lane position), physiological sensing (EEG, ECG) or camera-based facial analysis. Vehicle-dynamics methods react to behaviour that has already degraded [4]; physiological sensing is accurate but intrusive; a single RGB camera is passive and cheap.

4.2 PROBLEMS AND WEAKNESSES

Static thresholds cause bias between drivers with different eye shapes, binary outputs hide severity, and cloud designs depend on connectivity and expose face images.

4.3 REQUIREMENTS OF THE PROPOSED SYSTEM

- Functional: live landmark extraction, per-session calibration, four-level grading, on-screen display and audio alerts.
- Non-functional: CPU-only, around 30 FPS, fully offline, no image storage, runtime memory of about 800 MB.

4.4 SYSTEM FEASIBILITY

The system uses open-source tools (Python 3.11, MediaPipe Tasks, OpenCV, PyTorch, Pygame) and ran on a laptop CPU (Intel Core i7-7820HQ), so it is technically and economically feasible. Integration with vehicle systems and embedded targets is untested.

4.5 SELECTION OF TECHNIQUES

MediaPipe was selected for CPU-capable 3D landmarks, EAR/PERCLOS for their established link to alertness, and a BiGRU with attention for its small size (63,939 parameters, 0.25 MB) and interpretable attention weights.

CHAPTER 5 SYSTEM DESIGN AND METHODOLOGY

5.1 PIPELINE OVERVIEW

Figure 5.1 shows the five stages. Stages 1 to 3 and the display are implemented; stage 4 runs a Random Forest trained on synthetic data in the POC, while the production classifiers are specified in Section 5.5.

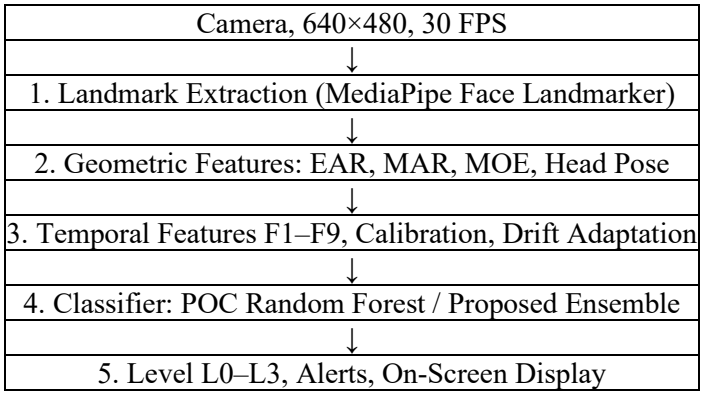


Fig 5.1 ALERT Processing Pipeline

5.2 LANDMARKS AND GEOMETRIC FEATURES

Frames are captured at 640×480 and 30 FPS with a one-frame buffer. Fixed landmark subsets are used: six points per eye (left 33, 160, 158, 133, 153, 144; right 362, 385, 387, 263, 373, 380), eight inner-lip points and six head-pose anchors (nose tip 1, chin 152, eye corners 33 and 263, mouth corners 61 and 291). With p1 to p6 the eye landmarks of [5]:

$$EAR = (||p2 - p6|| + ||p3 - p5||) / (2\,||p1 - p4||) \quad (5.2.1)$$

EAR is computed per eye and averaged. For yaw $|\psi| > 10°$ from solvePnP, a heuristic correction is applied (not validated against ground truth):

$$EARc = EAR / \cos\psi, \;\; \cos\psi > 0.1 \quad (5.2.2)$$

The mouth aspect ratio over eight inner-lip points and the combined signal are:

$$MAR = (||p2 - p8|| + ||p3 - p7|| + ||p4 - p6||) / (3\,||p1 - p5||), \;\; MOE = MAR / EAR \quad (5.2.3)$$

Head pose uses a six-point canonical 3D face model with OpenCV solvePnP, intrinsics approximated from the frame size (focal length equal to frame width) and zero distortion; pitch, yaw and roll follow from Rodrigues decomposition.

5.3 TEMPORAL FEATURES AND CALIBRATION

Raw EAR and MAR are smoothed with a 5-frame moving average. Table 5.1 lists the nine per-frame features.

Table 5.1 Feature Vector Computed Per Frame

ID	Feature	Window
F1	EAR (corrected, smoothed)	per frame
F2	EAR standard deviation	5 s
F3	PERCLOS-style closure ratio	60 s
F4	Blink rate	60 s
F5	Mean blink duration (ms)	60 s
F6	MAR (smoothed)	per frame
F7	Yawn count	5 min
F8	Head pitch (deg)	per frame
F9	MOE = MAR/EAR	per frame

5.3.1 PERCLOS-Style Closure Ratio

Reference PERCLOS counts time at 80% eyelid closure or more [4], [7]. ALERT instead counts the fraction of frames in a 60-second window with EAR below the calibrated threshold, which approximates the standard measure and is reported as such.

5.3.2 Calibration

At session start the driver looks forward for 10 s (300 frames). Frames with EAR < 0.12 are discarded to exclude blinks, and the thresholds are set as:

$$T_EAR = clamp(\mu E - 1.5\,\sigma E,\ 0.15,\ 0.30);\quad T_MAR = clamp(\mu M + 2.0\,\sigma M,\ 0.40,\ 0.70) \tag{5.3.1}$$

If σE > 0.06 the calibration is treated as noisy and defaults (T_EAR = 0.20, T_MAR = 0.50) are used. Thresholds are stored in a small local JSON file; no images are kept.

5.3.3 Drift Adaptation and Events

For long drives a slow exponential moving average updates the EAR baseline, using only frames with EAR > 1.2 T_EAR so closing eyes do not pull it down:

$$\mu E(t) = (1 - \alpha)\, \mu E(t-1) + \alpha\, EAR(t), \quad \alpha = 0.001 \tag{5.3.2}$$

At 30 FPS the time constant is about 33 minutes. A blink is a dip of EAR below threshold that recovers, with duration accepted between 100 and 800 ms (a blink typically lasts 100 to 400 ms [5], so the upper bound deliberately admits slow closures and should be revisited). A yawn is MAR above T_MAR for at least 2 s.

5.4 FATIGUE LEVELS

Table 5.2 gives the four levels. The bands are provisional engineering choices not derived from labelled data; tuning them is the first task for real-data experiments.

Table 5.2 Fatigue Levels and Provisional Bands

Level	Name	Closure ratio	Action
L0	Alert	< 0.08	Monitor
L1	Mild	0.08–0.15	Advisory chime
L2	Moderate	0.15–0.40	Repeated alarm
L3	Severe	> 0.40, or eyes closed > 1.5 s	Critical alarm

5.5 CLASSIFICATION

5.5.1 POC Classifier

A Random Forest (30 trees, maximum depth 8) with standard scaling, trained on synthetic data, outputs four class probabilities. A rule overrides the output to L3 when head pitch is below −25° or the closure ratio exceeds 50%.

5.5.2 Production Sequence Model

For RLDD (alert, low vigilant, drowsy [14]) a three-class head takes a 60-frame window of eight features through two stacked bidirectional GRU layers (64 and 32 units), a temporal attention layer and two dense layers (Figure 5.2). The model has 63,939 parameters and a

0.25 MB footprint, and inference runs every 15 frames (about 0.5 s at 30 FPS). Attention weights are drawn as a strip in the display. Because RLDD videos are below 30 FPS, windows should be defined in seconds and resampled.

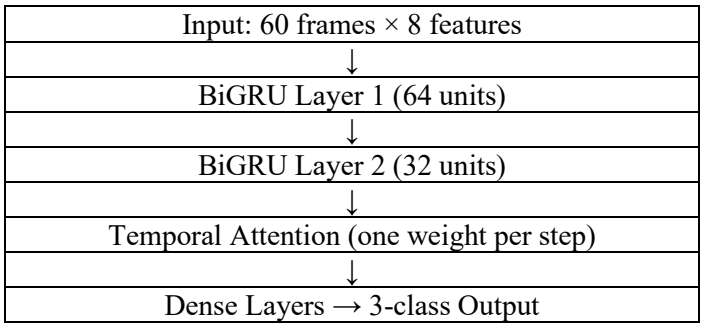


Fig 5.2 Production BiGRU Model Architecture

5.5.3 Proposed Ensemble

A second design combines three branches by weighted soft vote, with initial prior weights to be tuned on validation data:

$$P = 0.50\ P_LGBM + 0.30\ P_GRU + 0.20\ P_CNN \quad (5.5.1)$$

The branches are LightGBM [19] on the nine features, a 32-unit GRU over a 5-second window of F1, F3, F5 and F8, and a small CNN on 16×32 grayscale eye crops. L3 is forced if the LightGBM L3 probability exceeds 0.85 or the eyes stay closed for more than 45 consecutive frames. This is a design, not a measured improvement.

5.6 PLANNED TRAINING CONFIGURATION

Table 5.3 Planned Training Configuration

Item	**Setting**
Data	RLDD (primary), NTHU-DDD (robustness)
Split	Cross-subject folds
Optimiser	AdamW, lr 1e-3, weight decay 1e-4
Schedule	Cosine annealing, Tmax = 50 epochs
Loss, sampling	Cross-entropy, inverse-frequency sampler
Stopping	Patience 10 on validation loss; max 100 epochs
Batch size	64 (128 found best for CPU throughput in benchmarking)
Augmentation	Gaussian noise (0.01), frame dropout (10–20%), time scaling (±10%)

CHAPTER 6 IMPLEMENTATION AND RESULTS

6.1 IMPLEMENTATION ENVIRONMENT

Python 3.11, MediaPipe Tasks, OpenCV, Pygame, PyTorch 2.x (CPU, Intel MKL/OpenMP), NumPy and Git were used on an Intel Core i7-7820HQ laptop. Runtime memory is about 800 MB with MediaPipe, PyTorch and OpenCV loaded.

6.2 MODULES

- Feature extractor: landmarks, EAR, MAR, MOE and solvePnP head pose.
- Calibrator: 10-second adaptive baseline with noise rejection and drift adaptation.
- Model: two-stage BiGRU with temporal self-attention.
- Alert manager: normal, warning and critical state machine with procedural audio tones.
- Dashboard and overlay: live OpenCV HUD with state, confidence, feature values, class probabilities, attention strip and a warm-up bar until the 60-frame buffer fills.
- Data tools: RLDD loader with cross-subject folds, feature caching, augmentation (time_warp interpolation bug fixed), trainer and evaluation script.

6.3 FINDINGS AND OUTCOMES

All 141 RLDD videos were processed into 84,942 sliding-window sequences stored as a 29 MB compressed cache across four folds. DataLoader throughput reached 1,510 samples/s and a one-epoch CPU benchmark took 68.5 s. The live POC runs end to end with calibration and tiered audio alarms. The development log records about 25 ms per-frame latency; [replace with measured per-stage timings].

6.4 PROOF-OF-CONCEPT EVALUATION

The POC classifier is trained and tested on 10,000 synthetic draws (2,500 per class) from class-conditional Gaussians (Table 6.1) with an 80/20 stratified split. Because classes are separated by construction, high accuracy here only shows that the pipeline runs end to end and says nothing about detecting fatigue in real drivers. Accuracy on this split is therefore not presented as a result. [Insert per-class precision, recall and F1 from POC_v0/train_model.py for completeness.]

Table 6.1 Synthetic Class Parameters (Mean, Std)

Class	**EAR**	**Closure (%)**	**Blink dur. (ms)**
L0	0.30, 0.025	5.0, 3.0	200, 30
L1	0.25, 0.020	20.0, 2.5	300, 40
L2	0.20, 0.018	32.0, 3.5	420, 50
L3	0.14, 0.020	55.0, 8.0	650, 120

6.5 RESULT ANALYSIS

Under a Gaussian assumption a threshold at $\mu - 1.5\sigma$ lies below about 93% of open-eye samples, so about 7% of open-eye frames would count as closed, a baseline closure ratio near the 0.08 L0/L1 boundary. Real EAR distributions are not Gaussian and the blink filter removes some low samples, but this interaction must be measured and the multiplier or bands adjusted. The synthetic generator also disagrees with the provisional bands (the L1 mean of 20% falls in the L2 band of Table 5.2); one should be changed before further tuning.

CHAPTER 7 TESTING

7.1 TESTING STRATEGY

Testing so far is functional, at the level of modules and the end-to-end POC. No accuracy testing on real footage has been performed; this is the main gap.

7.2 TEST CASES AND RESULTS

Table 7.1 Test Cases

ID	Test condition	Expected output	Actual output / remark
T1	Calibration on synthetic landmark stream	Completes at frame 300	Completed at expected frame
T2	Calibration with $\sigma E > 0.06$	Fallback to default thresholds	[Verify and record]
T3	Live webcam run	Overlay, level and audio alerts shown	POC ran end to end
T4	Feature cache over 141 RLDD videos	84,942 windows, 4 folds	Verified
T5	Time-warp augmentation	No array-size mismatch	Fixed and verified
T6	CPU training, 1 epoch	Completes, throughput logged	68.5 s per epoch
T7	RLDD cross-subject evaluation	Per-class F1, confusion matrix	Pending
T8	Glasses and low light	Graceful degradation	Pending

CHAPTER 8 CONCLUSION AND DISCUSSION

8.1 SUMMARY OF PROJECT WORK

ALERT is an offline fatigue monitor built from MediaPipe landmarks, a nine-feature temporal vector, per-session calibration and a four-level grade. The POC works end to end, and the RLDD feature cache and CPU training benchmark are complete. The production sequence model and the proposed ensemble are specified and await training, so no claim about real-world accuracy is made.

8.2 PROBLEMS ENCOUNTERED AND SOLUTIONS

- MediaPipe Tasks API integration without legacy solutions: handled by building on the Tasks API directly.
- Linux camera index (loopback /dev/video0 versus hardware /dev/video1): camera index selection corrected.
- Long CPU extraction (about 3.5 hours) with a silent terminal: logging fixed.

8.3 LIMITATIONS

- No real-data results; thresholds, level bands, the 800 ms blink bound, the yaw correction and the −25° pitch override are untested heuristics.
- The ensemble needs an ablation against its best single branch to justify its cost.
- Eyewear and low light degrade landmarks; remedies include CLAHE [20], the CNN branch and a near-infrared path needing recalibration.
- Deployment has only been on a laptop CPU; ONNX Runtime, OpenVINO or TensorRT export is untested; haptic output needs hardware.

8.4 FUTURE ENHANCEMENT

Next milestones are RLDD cross-subject training compared with the 65.2% baseline [14] and the 86.52% ensemble result [16], tuning of level bands and the calibration rule, measured latency, ONNX export and tests under glasses and low light.

REFERENCES

[1] National Highway Traffic Safety Administration, “Drowsy driving.” [Online]. Available: https://www.nhtsa.gov/risky-driving/drowsy-driving. Accessed: Oct. 23, 2026.

[2] AAA Foundation for Traffic Safety, “Drowsy driving in fatal crashes, United States, 2017–2021,” Research Brief, Mar. 2024. [Online]. Available: https://www.aaafoundation.org/drowsy-driving

[3] European Parliament and Council, “Regulation (EU) 2019/2144 on type-approval requirements for motor vehicles … as regards their general safety,” Official J. Eur. Union, Nov. 27, 2019. [Online]. Available: https://eur-lex.europa.eu/eli/reg/2019/2144/eng

[4] W. W. Wierwille, S. S. Wreggit, C. L. Kirn, L. A. Ellsworth, and R. J. Fairbanks, “Research on vehicle-based driver status/performance monitoring,” NHTSA, Washington, DC, USA, Rep. DOT HS 808 247, Dec. 1994.

[5] T. Soukupová and J. Čech, “Real-time eye blink detection using facial landmarks,” in Proc. 21st Comput. Vis. Winter Workshop (CVWW), Rimske Toplice, Slovenia, Feb. 2016.

[6] D. F. Dinges, M. M. Mallis, G. Maislin, and J. W. Powell IV, “Evaluation of techniques for ocular measurement as an index of fatigue and as the basis for alertness management,” NHTSA, Rep. DOT HS 808 762, Apr. 1998.

[7] D. F. Dinges and R. Grace, “PERCLOS: A valid psychophysiological measure of alertness as assessed by psychomotor vigilance,” FHWA Tech Brief FHWA-MCRT-98-006, Oct. 1998.

[8] T. Abe, “PERCLOS-based technologies for detecting drowsiness: Current evidence and future directions,” Sleep Adv., vol. 4, no. 1, Art. no. zpad006, Jan. 2023, doi: 10.1093/sleepadvances/zpad006.

[9] V. Kazemi and J. Sullivan, “One millisecond face alignment with an ensemble of regression trees,” in Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR), 2014, pp. 1867–1874.

[10] C. Lugaresi et al., “MediaPipe: A framework for building perception pipelines,” 2019, arXiv:1906.08172.

[11] Y. Kartynnik, A. Ablavatski, I. Grishchenko, and M. Grundmann, “Real-time facial surface geometry from monocular video on mobile GPUs,” 2019, arXiv:1907.06724.

[12] Google, “Face landmark detection guide,” MediaPipe Solutions. [Online]. Available: https://developers.google.com/edge/mediapipe/solutions/vision/face_landmarker. Accessed: Oct. 23, 2026.

[13] C.-H. Weng, Y.-H. Lai, and S.-H. Lai, “Driver drowsiness detection via a hierarchical temporal deep belief network,” in Computer Vision – ACCV 2016 Workshops, LNCS vol. 10118. Cham, Switzerland: Springer, 2017, pp. 117–133.

[14] R. Ghoddoosian, M. Galib, and V. Athitsos, “A realistic dataset and baseline temporal model for early drowsiness detection,” in Proc. IEEE/CVF CVPRW, 2019, pp. 178–187, doi: 10.1109/CVPRW.2019.00027.

[15] S. Park, F. Pan, S. Kang, and C. D. Yoo, “Driver drowsiness detection system based on feature representation learning using various deep networks,” in Computer Vision – ACCV 2016 Workshops, LNCS vol. 10118. Springer, 2017, pp. 154–164.

[16] C. Xu, W. Huang, J. Liu, and L. Li, “Detecting driver drowsiness using hybrid facial features and ensemble learning,” Information, vol. 16, no. 4, Art. no. 294, Apr. 2025, doi: 10.3390/info16040294.

[17] K. Cho et al., “Learning phrase representations using RNN encoder–decoder for statistical machine translation,” in Proc. EMNLP, 2014, pp. 1724–1734.

[18] D. Bahdanau, K. Cho, and Y. Bengio, “Neural machine translation by jointly learning to align and translate,” in Proc. ICLR, 2015.

[19] G. Ke et al., “LightGBM: A highly efficient gradient boosting decision tree,” in Adv. Neural Inf. Process. Syst. (NeurIPS), vol. 30, 2017, pp. 3146–3154.

[20] S. M. Pizer et al., “Adaptive histogram equalization and its variations,” Comput. Vis., Graph., Image Process., vol. 39, no. 3, pp. 355–368, 1987.